

RESTful Web Services



A brief introduction.

Dozent: Prof. Dr. Michael Eichberg
Kontakt: michael.eichberg@dhbw.de
Version: 1.0
Quelle: (teilweise) RESTful Web Services; Leonard Richardson & Sam Ruby; O'Reilly

Slides/Script: https://delors.github.io/ds-introduction_to_rest/folien.en.rst.html
https://delors.github.io/ds-introduction_to_rest/folien.en.rst.html.pdf
Reporting errors: <https://github.com/Delors/delors.github.io/issues>

What is a *Web Service* in the context of RESTful Web Services?

Traditional view

A *Web Service* is simply a web page that can be requested and processed by a computer.

A *Web Service* is a "web page" that is to be consumed by an *autonomous programme* - as opposed to a web browser or similar UI tool.



REST^[1]

- REST = Representational State Transfer

(Essentially a set of design principles for judging architecture; **an architectural style**).

- Resources are identified by uniform resource identifiers (URIs)
- Resources are manipulated by their representations
- Messages are self-describing and stateless

Of secondary importance:

- Multiple representations are accepted or sent
- "Hypertext" represents the application state

[1] REST was described by Roy Fielding in his dissertation.

A possible architecture for RESTful web services

Resource-oriented Architecture (ROA)

- Information about the method is included in the HTTP method.
- Scoping information is included in the URI.

(I. e. which data is affected.)

REST-Style

- Client-server
- stateless
- Cached
- Uniform Interface (HTTP Methods)
- Multi-layered system

RESTful Web Services - Foundations

HTTP: the underlying stateless transport protocol:
Essential methods:

GET: sideeffect-free requests for information

POST: adding new information (without specifying the target URI)

PUT: idempotent update or creation of new information at the given URI

DELETE: idempotent deletion of information

URI: used to find resources

Representation: **JSON**, XML, SVG, WebP, XML, ...

Two Types of State

1

Application State / Session State

- "State" refers to Application-/Session State

The application state is the information necessary to understand the context of an interaction

Authorization and authentication information are examples of application state.

- Maintained as part of the content transmitted from the client to the server and back to the client. I. e. the client manages the application state.
- Thus, any server can potentially resume the transaction at the point where it was interrupted.

2

Resource State

- The resource state is the type of state that the *S* in *REST* refers to.
- The *stateless* restriction means that all messages must contain the entire application state (i. e. we effectively have no sessions).

Multiple representations

- Most resources only have a single representation.
- REST can support any media type; JSON is the standard.
(HTTP supports content negotiation.)
- Links can be embedded and reflect the structure with which a user can navigate through an application.

Simple/first tests for RESTfulness

- Can I use a GET to retrieve the URLs I have POSTed to?
- Would the client notice if the server...
 - is restarted at any point between requests
 - is reinitialized when the client makes the next request.

Resource modelling

- organize the application into URI-addressable resources (discrete resources should have their own stable URIs).
 - use only the standard HTTP messages - GET, PUT, POST, DELETE and PATCH - to provide the full capabilities of the application
-

HTTP methods

GET is used to query resources.

PUT is used to create a resource or update it if you know the URI.

POST is used to create a new resource. The response should then contain the URI of the created resource.

DELETE deletes the specified resource.

The difference between **PUT** and **POST** is that **PUT** is idempotent: a single or repeated calls have the same effect (i. e. a repeated call has no side effect), while successive identical **POST** calls can have additional effects, such as the repeated transfer of an order/the repeated creation of a message.

A **PATCH** request is regarded as a set of instructions for changing a resource. In contrast, a PUT request is a complete representation of a resource.

Example Application del.icio.us

Quelle: <https://www.peej.co.uk/articles/restfully-delicious.html>

del.icio.us enables us:

- to get a list of all our bookmarks and filter this list by tags or date and to limit the number of retrieved bookmarks
- to retrieve the number of bookmarks created on different days
- to retrieve when we last updated our bookmarks
- to retrieve a list of all our markers
- to add a bookmark
- to edit a bookmark
- to delete a bookmark
- to rename a bookmark

Example Application del.icio.us: Resources

Bookmarks: *[http://del.icio.us/api/\[username\]/bookmarks](http://del.icio.us/api/[username]/bookmarks)*

Tags: *[http://del.icio.us/api/\[username\]/tags](http://del.icio.us/api/[username]/tags)*

[username]: is the username of the user whose bookmarks we are interested in

Example Application del.icio.us: Repräsentation von Ressourcen

We define (in this example) some XML document formats and media types to identify them:

Mediatype	Description
delicious/bookmarks+xml	list of bookmarks
delicious/bookmark+xml	one bookmark
delicious/bookmarkcount+xml	number of bookmarks per tag
delicious/update+xml	time at which the bookmarks were last updated
delicious/tags+xml	list of tags
delicious/tag+xml	a tag

Example Application del.icio.us: Query Bookmarks

URL: *http://del.icio.us/api/[username]/bookmarks/*

Method: GET

Querystring: tag = Filter by tag

dt = Filter by date

start = The number of the first returned bookmark

end = The number of the last returned bookmark

Return value: 200 OK & XML (delicious/bookmarks+xml)

401 Unauthorized

404 Not Found

Example application del.icio.us: Query bookmarks - example response

GET <http://del.icio.us/api/peej/bookmarks/?start=1&end=2>

```
1 <?xml version="1.0"?>
2 <bookmarks start="1" end="2"
3   next="http://del.icio.us/api/peej/bookmarks?start=3&end=4">
4   <bookmark url="http://www.example.org/one" tags="example,test"
5     href="http://del.icio.us/api/peej/bookmarks/a211528fb5108cddaa4b0d3aeccdbdcf"
6     time="2005-10-21T19:07:30Z">
7     Example of a Delicious bookmark
8   </bookmark>
9   <bookmark url="http://www.example.org/two" tags="example,test"
10    href="http://del.icio.us/api/peej/bookmarks/e47d06a59309774edab56813438bd3ce"
11    time="2005-10-21T19:34:16Z">
12    Another example of a Delicious bookmark
13  </bookmark>
14 </bookmarks>
```

Example application del.icio.us: Information about a bookmark

URL: *http://del.icio.us/api/[username]/bookmarks/[hash]`*
Method: *GET*
Return value: 200 OK & XML (delicious/bookmark+xml)
401 Unauthorized
404 Not Found

Example application del.icio.us: Information about a bookmark - Example response

GET <http://del.icio.us/api/peej/bookmarks/a211528fb5108cdd>

```
1 <?xml version="1.0"?>
2 <bookmark url="http://www.example.org/one" time="2005-10-21T19:07:30Z">
3   <description>
4     Example of a Delicious bookmark
5   </description>
6   <tags count="2">
7     <tag name="example" href="http://del.icio.us/api/peej/tags/example" />
8     <tag name="test" href="http://del.icio.us/api/peej/tags/test" />
9   </tags>
10 </bookmark>
```


Example application del.icio.us: Query the number of bookmarks

URL: *http://del.icio.us/api/[username]/bookmarks/count*

Method: GET

Query parameter: tag = Filter by tag

Return value: 200 OK & XML (delicious/bookmark+xml)

401 Unauthorized

404 Not Found

Example application del.icio.us: Query when the last change was made

URL: *http://del.icio.us/api/[username]/bookmarks/update*

Method: GET

Return value: 200 OK & XML (delicious/bookmark+xml) 401 Unauthorized 404
Not Found

Example application del.icio.us: Adding a bookmark

URL: *http://del.icio.us/api/[username]/bookmarks/`*

Method: POST

Query document: XML (delicious/bookmark+xml)

Return value: 201 Created & Location

401 Unauthorized

415 Unsupported Media Type(if the send document is not valid)

Example application del.icio.us: Adding a bookmark - example document

POST <http://del.icio.us/api/peej/bookmarks/>

```
1 <?xml version="1.0"?>
2 <bookmark url="http://www.example.org/one"
3   time="2005-10-21T19:07:30Z">
4   <description>Example of a Delicious bookmark</description>
5   <tags>
6     <tag name="example" />
7     <tag name="test" />
8   </tags>
9 </bookmark>
```

Example application del.icio.us: Update a bookmark

URL: *http://del.icio.us/api/[username]/bookmarks/[hash]`*

Method: PUT

Query document: XML (delicious/bookmark+xml)

Return value: 201 Created & Location

401 Unauthorized

404 Not Found (new bookmarks cannot be created using put!)

415 Unsupported Media Type (if the send document is not valid)

Example application del.icio.us: Delete a bookmark

URL: *http://del.icio.us/api/[username]/bookmarks/[hash]*
Method: DELETE
Return value: 204 No Content
401 Unauthorized
404 Not Found